

Coverage audit: the fragment and selected wider comparisons

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1 WHAT THIS AUDIT IS

The article's §5 compares four implementations of the same facts: the D-noun categorization with ordinary Head (I₁), a separate determinative category with ordinary Head (I₂), a separate category with *CGEL*'s fusion of functions (I₃), and the D-noun categorization with fusion (I₄). It states which conditions the accounts share and which are specific to one of them. This audit records how each implementation covers the construction families and lexeme groups in the fragment and selected wider comparisons: by a general rule of that grammar (D), by a general rule that applies only because the grammar admits a determinative as the lexical head of Nom and NP (D_n), by a rule the grammar has to state once for nouns and again for determinatives, or once for NP and again for DP (D₂), by a fact stored on the lexeme (L), by a condition stated for that construction alone (S), or not at all (U).

The classifications are the author's, made by hand from the article's stated conditions and the sections cited beside each cell; they are the analysis, not data. The precedence rule is this: a cell records the mechanism by which the implementation covers the licensed use. Shared lexical permissions, target selection, forms, and inflection are held fixed across the four implementations. A lexical fact is coded L where it's a permission or exclusion alone; where a licensed use is covered by a rule, the rule's class is recorded even though the permission is presupposed.

S and D₂ codes count cells; the distinct conditions and duplicated rules behind them are listed separately (Table 5), since a condition stated once may apply in several cells. The comparison concerns the rules and facts each account has to state for the audited constructions. The minimum description length principle motivates assessing complete representations (Rissanen, 1978), but this audit doesn't define a common encoding or estimate an MDL score.

Beside each cell the audit prints the number of participation claims the claim register records for that family and group, as licensed/conditional/excluded, so that a cell's classification can be checked against what the article asserts there. Cells with no claim and no analysis in the article are left out; the article's analysis, not the register, decides which cells exist.

2 THE FOUR IMPLEMENTATIONS AND THEIR PRIMITIVES

The classes count rules, not primitives. The primitives differ too, and Table 1 lists them, since a rule that is general in one grammar may be general only because that grammar has a primitive the other lacks.

Table 1: Primitives of the four specified implementations.

	I ₁	I ₂	I ₃	I ₄
Lexical categories	Noun, four subcategories	Noun (three) and D	Noun (three) and D	Noun, four subcategories
Category condition on words directly heading Nom	N	N or D	N	N
Phrase types in Det	NP, PP	NP, PP	DP, NP, PP	NP, PP
Head configurations for independent determinatives	ordinary Head	ordinary Head	Det–Head and Mod–Head fusion	Det–Head and Mod–Head fusion
Adjectival Mod–Head fusion (<i>the rich</i>)	yes	yes	yes	yes
Restrictions stated on the fused inner phrase	none	none	yes	yes
Complex determinatives listed	none	none	<i>a few, a little, many a, a great many</i>	none
Conversion rule for the set-denoting use	none	yes	yes	none
Domain of the head-genitive definition	Noun heads	Noun and D heads	Noun and D heads	Noun heads

In I₃, a determinative heads a DP, which fills Head in the further nominal structure; D doesn't directly head Nom. The category condition on words directly heading Nom is therefore N, as in I₁ and I₄.

3 RESULTS

Table 2: Cells by class, over the 103 classified cells.

Implementation	D	Dn	D ₂	L	S	U
I ₁ : D-noun, ordinary Head	61	0	0	36	6	0
I ₂ : Separate D, ordinary Head	28	32	1	36	6	0
I ₃ : Separate D, fused Head	40	0	12	35	16	0
I ₄ : D-noun, fused Head	53	0	0	36	14	0

Table 3: Cells on which the implementations differ.

Family	Lexeme group	I ₁	I ₂	I ₃	I ₄
independent argument use, including the existential	no/none	D	Dn	D	D

Family	Lexeme group	I ₁	I ₂	I ₃	I ₄
independent argument use, including the existential	demonstratives, such, and the wh-forms	D	Dn	D	D
independent argument use, including the existential	the general quantifiers	D	Dn	D	D
independent argument use, including the existential	the gradable four	D	Dn	D	D
independent argument use, including the existential	cardinals	D	Dn	D	D
independent argument use, including the existential	compound determinatives	D	Dn	D	D
independent use with a partitive of-PP	no/none	D	Dn	S	S
independent use with a partitive of-PP	the general quantifiers	D	Dn	S	S
independent use with a partitive of-PP	the gradable four	D	Dn	S	S
independent use with a partitive of-PP	cardinals	D	Dn	S	S
external determination of an independent use	the gradable four	D	Dn	D	D
external determination of an independent use	cardinals	D	Dn	D	D
internal modifiers admitted	the gradable four	D	Dn	D	D
peripheral modifiers admitted	every	D	Dn	D ₂	D
peripheral modifiers admitted	no/none	D	Dn	D ₂	D
peripheral modifiers admitted	the general quantifiers	D	Dn	D ₂	D
peripheral modifiers admitted	cardinals	D	Dn	D ₂	D
peripheral modifiers admitted	compound determinatives	D	Dn	D ₂	D
relative postmodification	demonstratives, such, and the wh-forms	D	Dn	S	S
relative postmodification	the general quantifiers	D	Dn	S	S
relative postmodification	the gradable four	D	Dn	S	S
relative postmodification	cardinals	D	Dn	S	S
relative postmodification	compound determinatives	D	Dn	S	S
number agreement and transparency	my/mine	S	S	D	D
number agreement and transparency	the general quantifiers	D	Dn	D ₂	D
number agreement and transparency	the gradable four	D	Dn	D ₂	D
genitive marking	compound determinatives	D	D ₂	D ₂	D
degree modification outside the NP; comparative complements; predicative use	the articles	D	Dn	D ₂	D

Family	Lexeme group	I ₁	I ₂	I ₃	I ₄
degree modification outside the NP; comparative complements; predicative use	no/none	D	D _n	D ₂	D
degree modification outside the NP; comparative complements; predicative use	the general quantifiers	D	D _n	D ₂	D
degree modification outside the NP; comparative complements; predicative use	the gradable four	D	D _n	D ₂	D
predeterminer and restricted peripheral modification; the former complex determinatives	the articles	L	L	S	L
predeterminer and restricted peripheral modification; the former complex determinatives	demonstratives, such, and the wh-forms	D	D _n	D	D
predeterminer and restricted peripheral modification; the former complex determinatives	the general quantifiers	D	D _n	D	D
predeterminer and restricted peripheral modification; the former complex determinatives	the gradable four	D	D _n	S	D

Table 4: What each implementation states for a construction alone (S) or states twice (D2).

Account	Stated for the construction alone	Stated twice
I1	internal modifiers (cardinals); the compound construction (compound determinatives); number agreement (my/mine); the secondary set-denoting use (cardinals, compound determinatives, proper nouns)	–
I2	internal modifiers (cardinals); the compound construction (compound determinatives); number agreement (my/mine); the secondary set-denoting use (cardinals, compound determinatives, proper nouns)	genitive marking (compound determinatives)
I3	partitive of-PP (no/none, the general quantifiers, the gradable four, cardinals); internal modifiers (cardinals); relative postmodification (demonstratives, such, and the wh-forms, the general quantifiers, the gradable four, cardinals, compound determinatives); the compound construction (compound determinatives); predeterminer and restricted peripheral modification (the articles, the gradable four); the secondary set-denoting use (cardinals, compound determinatives, proper nouns)	peripheral modifiers (every, no/none, the general quantifiers, cardinals, compound determinatives); number agreement (the general quantifiers, the gradable four); genitive marking (compound determinatives); degree use outside the NP (the articles, no/none, the general quantifiers, the gradable four)
I4	partitive of-PP (no/none, the general quantifiers, the gradable four, cardinals); internal modifiers (cardinals); relative postmodification (demonstratives, such, and the wh-forms, the general quantifiers, the gradable four, cardinals, compound determinatives); the compound construction (compound determinatives); the secondary set-denoting use (cardinals, compound determinatives, proper nouns)	–

Table 5: Distinct conditions stated for a construction alone, and distinct rules stated twice, per implementation. One condition may cover several cells.

Implementation	Conditions	Rules stated twice	The conditions
I ₁	4	0	cardinal two sites, compound restrictor, mine agreement, secondary use rule
I ₂	5	1	cardinal two sites, compound restrictor, conversion rule, mine agreement, secondary use rule
I ₃	6	4	cardinal two sites, complex listing, compound restrictor, conversion rule, inner phrase restriction, secondary use rule
I ₄	4	0	cardinal two sites, compound restrictor, inner phrase restriction, secondary use rule

Table 6: Sensitivity under alternative coding conventions: lexical licensing takes precedence; the fused inner-phrase restriction is counted once; typed schemas replace duplicated rules; and the secondary use is construction-specific in every account.

Implementation	D	Dn	D2	L	S	U
I ₁	54	0	0	43	6	0
I ₂	29	25	0	43	6	0
I ₃	54	0	0	42	7	0
I ₄	55	0	0	43	5	0

Table 7: Sensitivity under the same alternative coding conventions, additionally treating the N or D head condition as a general rule (Dn recoded as D).

Implementation	D	Dn	D2	L	S	U
I ₁	54	0	0	43	6	0
I ₂	54	0	0	43	6	0
I ₃	54	0	0	42	7	0
I ₄	55	0	0	43	5	0

Fusion is held fixed for adjectives (*the rich*) in all four implementations. *CGEL* (I₃) uses it for independent determinatives too, with one condition on the fused inner phrase (nine cells) and rules stated for NPs and DPs separately (twelve cells, four rules). Both ordinary-Head implementations remove this use. I₁ reclassifies determinatives within Noun; I₂ admits N or D as lexical Head in Nom. Its 32 Dn cells record uses of that one condition.

The stated I₂ also retains conversion to Noun in the secondary use and extends the head-genitive definition to D. Those choices aren't obligatory in every separate-D formulation. A nominal feature could be assigned once to Noun and once to D, with inheritance within each category. Table 8 com-

pares how that alternative and I₁ express the shared grammar, holding the lexical restrictions and input-specific interpretations fixed.

Table 8: Shared declarations under I₁ and a separate-D feature formulation. The latter isn't separately coded in the grid.

Declaration	I ₁	Separate D with a nominal feature
Shared domain	Noun, with four subcategories	Noun (three subcategories) and D each bear the feature; their members inherit it
Lexical Head in Nom	Noun	Bearer of the nominal feature
Head-genitive definition	One definition over Noun heads	One definition over heads bearing the feature
Secondary use	One schema with the three stated inputs; primary Noun retained	One schema with the same inputs; D inputs change category if <i>CGEL</i> 's output categories are retained

The feature's extension matches that of the broader Noun category. It permits shared conditions without assignments to individual lexemes or a separate semantic process for each primary category. I₁ expresses this domain through the Noun category supported by the article's profile comparison; the alternative retains the primary boundary and supplies a shared feature. A strict description-length ranking would require a common encoding and costs for both representations. The present audit doesn't supply that ranking.

For the former complex determinatives, I₁, I₂, and I₄ state a restricted peripheral-modifier permission for *a*, with head-specific modifier selection and ordering; I₃ lists the complex expressions. This permission is recorded in the article cell, rather than treated as an ordinary Det use of *a*. Positive polarity in *a few* and *a little* remains a convention of the construction under the decomposition; the cell totals don't measure the complexity of that interpretation.

The sensitivity tables apply alternative coding conventions: lexical licensing first, the inner-phrase restriction counted once, typed schemas in place of duplicated rules, and the secondary use treated as a construction-specific rule in every account. Under those conventions I₁, I₃, and I₄ fall within one cell of each other; recoding D_n as D also brings I₂ level with I₁. This shows the dependence of the cell totals on the coding. Table 5 inventories the listed conditions and duplicated rules, not the cost of category nodes or features. The feature alternative makes sharing explicit without treating its definition as costless.

Table 9: The grid. Codes: D derived by a general rule; Dn derived only by admitting D as the head of Nom and NP; D2 derived by a rule stated once for nouns and again for determinatives (or for NP and DP); L stored on the lexeme; S stated for this construction alone; U not covered. Claims: register claims for the cell, as licensed/conditional/excluded.

Lexeme group	I1	I2	I3	I4	Claims	Basis
<i>Det before a nominal (§2.4)</i>						
the articles	L	L	L	L	5/2/1	Det use permission and target selection (a: singular count) are lexical in every account; the phrase in Det is an NP under I1, I2, and I4 and a DP under I3 (primitives table).
every	L	L	L	L	4/1/1	as the articles; the Mod use after a genitive Det is a separate lexical permission (internal Mod family).
no/none	L	L	L	L	1/1/1	the dependent form no is selected by the Det use (forms family).
my/mine	L	L	L	L	1/2/1	a genitive NP in Det, the general realization; the my/mine form selection is lexical (forms family).
demonstratives, such, and the wh-forms	L	L	L	L	4/1/0	Det use permission; agreement with the target is the demonstratives' inflection (inflection family).
the general quantifiers	L	L	L	L	8/1/1	Det use permission with target restrictions (some: plural or non-count; each: singular count).
the gradable four	L	L	L	L	4/1/1	Det use permission with target restrictions (few: plural count; much: non-count).
cardinals	L	L	L	L	3/1/0	Det use with plural targets above one; complex cardinals fill Det as phrases (§5.4).
compound determinatives	D	D	D	D	1/0/0	only as genitive NPs (someone's preferences): the general genitive-Det realization.
pronouns	D	D	D	D	0/0/2	genitive NPs (my preferences, her preferences) fill Det by the general genitive-Det realization; plain-case pronouns don't.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	2/0/0	genitive NPs (people's preferences) by the general realization; plain-case NPs (what size shoes, Sunday morning) by a restricted construction that every account states separately.
proper nouns	D	D	D	D	5/0/0	genitive NPs (Kim's preferences); Sunday morning is the restricted plain-case construction.
<i>internal Mod before a nominal (§4.3)</i>						
every	L	L	L	L	1/1/0	Mod after a genitive Det (her every move): a lexical permission.
the general quantifiers	L	L	L	L	1/0/1	several, certain, various as modifiers (its several advantages): lexical.
the gradable four	L	L	L	L	4/1/0	the few people, her many virtues: a lexical permission for Mod use; CGEL gives few Mod after a definite determiner (p. 392).
cardinals	L	L	L	L	2/0/0	these two hundred books, the remaining three: Mod use of cardinals.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	2/o/o	nominal premodifiers (dog houses): general.
proper nouns	D	D	D	D	1/o/o	Canada Day: general nominal premodification, restricted for names.
adjective controls	D	D	D	D	4/o/o	attributive AdjPs: general.
<i>independent argument use, including the existential (§4.2)</i>						
the articles	L	L	L	L	1/1/6	no independent argument use: the lexeme lacks the use permission in every account.
every	L	L	L	L	0/1/4	The fragment represents every's ordinary dependent pattern in all four accounts. The saved NOW material contains anaphoric occurrences (two confirmed on publication pages, one source variant); those lie outside this fragment. This cell is not an empirical absence claim; see the quantifier supplement.
no/none	D	Dn	D	D	4/1/2	none as the independent form (forms family); the use is licensed by the ordinary Head chain (I ₁ , I ₂) or by Det–Head fusion (I ₃ , I ₄). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
my/mine	D	D	D	D	2/o/2	mine: ordinary Head (I ₁ , I ₂) or fusion with an understood nominal (I ₃ , I ₄); the agreement condition is in the agreement family.
demonstratives, such, and the wh-forms	D	Dn	D	D	6/o/1	independent this/that; which and what as interrogative heads. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	Dn	D	D	17/8/o	some, all, both, each, several, enough independently, and the existential There are several: ordinary Head (I ₁ , I ₂) or Det–Head fusion (I ₃ , I ₄). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	Dn	D	D	10/3/o	few and many independently, including bare few with no set under discussion (§4.2). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
cardinals	D	Dn	D	D	1/o/o	independent two, ten; the almost thirty who came is in the external-determination family. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
compound determinatives	D	Dn	D	D	4/o/o	someone, anyone as NPs; under I ₃ the DP–Nom structure of Payne 2007. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
pronouns	D	D	D	D	11/o/2	the noun-headed NP without a determiner (She left).
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	6/1/1	bare plurals and non-counts; singular count arguments require determination, a condition every account states.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
proper nouns	D	D	D	D	5/0/0	primary naming uses (Kim left).
adjective	D	D	D	D	8/4/0	the rich by adjectival Mod–Head fusion, retained in all four; bare numerous, multiple, countless are attested overlaps the article leaves unanalysed (§2.5), so a U would be right for them alone.
controls						
<i>independent use with a partitive of-PP (§4.3)</i>						
every	L	L	L	L	0/0/1	no partitive use (each of, not every of): lexical.
no/none	D	Dn	S	S	0/2/1	none of the students: the of-PP is a complement in the Nom that none heads (I ₁ , I ₂); under fusion it attaches outside the DP, which the inner-phrase restriction has to state (§5.2). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	Dn	S	S	3/3/0	some of the wine, many of them: complement in the Nom the determinative heads (I ₁ , I ₂); outside the fused DP by a stated restriction (I ₃ , I ₄), §5.2. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	Dn	S	S	2/1/1	few of them, many of the delegates: as for the general quantifiers. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
cardinals	D	Dn	S	S	–	two of them: as above. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	4/0/0	a lot of the wine, a number of protesters: complement of the head noun in every account (CGEL's own analysis of a number of, pp. 351–352).
<i>external determination of an independent use (§4.3)</i>						
the articles	L	L	L	L	1/0/0	excluded: the permissions of independent few don't transfer to the articles (§5.2).
every	L	L	L	L	–	excluded, as the articles.
the general quantifiers	L	L	L	L	–	excluded (the some, the all): lexical.
the gradable four	D	Dn	D	D	3/2/0	the few, the lucky few: Det plus a Nom headed by few (I ₁ , I ₂); Mod–Head fusion, the adjectival configuration (I ₃ , I ₄), with the exclusion of a determiner inside the fused phrase stated among the inner-phrase restrictions (§5.2). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
cardinals	D	Dn	D	D	0/1/0	the two, these three, the almost thirty who came: as for few. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
compound determinatives	L	L	L	L	–	excluded as determinatives by the compound construction; the set-denoting a special someone is the secondary use.
pronouns	L	L	L	L	1/0/0	normally excluded; poor old me is a restricted case.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	o/1/o	the general case.
proper nouns	L	L	L	L	–	restricted in primary naming uses; the Smiths and an Ophelia are the secondary use.
adjective controls	D	D	D	D	–	the rich, the idle rich: adjectival Mod–Head fusion in all four.
<i>internal modifiers admitted (§4.4)</i>						
the general quantifiers	L	L	L	L	1/o/2	degree modifiers excluded for the non-gradable determinatives (very some): lexical selection.
the gradable four	D	Dn	D	D	1o/o/o	degree AdvPs select the gradable four (a lexical selection) and attach inside the Nom (I ₁ , I ₂) or inside the DP (I ₃ , I ₄); one permission covers the very few people and the very few who objected in every account. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
cardinals	S	S	S	S	–	approximatives attach internally after an external determiner (the almost thirty who came) and peripherally otherwise (almost ten): a two-site condition every account states (§4.4).
pronouns	L	L	L	L	3/o/1	poor old me: restricted AdjP premodification.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	o/1/1	productive AdjP and nominal premodification.
proper nouns	L	L	L	L	–	restricted premodification (architect Norman Foster).
adjective controls	D	D	D	D	3/o/1	very tall: degree modification of adjectives.
<i>peripheral modifiers admitted (§4.4)</i>						
every	D	Dn	D ₂	D	4/o/o	almost every: NP-peripheral attachment by the general rule for peripheral modifiers, with lexical selection (I ₁ , I ₂ , I ₄); a modifier inside the DP under CGEL, so the same fact needs the DP-modifier rule beside the NP-peripheral rule that covers only you (I ₃), §4.4. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
no/none	D	Dn	D ₂	D	3/o/o	practically no: as for every. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	Dn	D ₂	D	5/o/o	nearly all, hardly any: as for every. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
cardinals	D	Dn	D ₂	D	1/o/o	almost ten: as for every. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
compound determinatives	D	Dn	D ₂	D	–	hardly anyone: peripheral to the NP that anyone heads (I ₁ , I ₂ , I ₄); in DP structure under Payne 2007 (I ₃). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
pronouns	D	D	D	D	2/o/o	only you, all we who signed up: NP-peripheral in every account.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	–	only the book, almost the whole class.
proper nouns	D	D	D	D	–	even Kim.
<i>relative postmodification (§2.4)</i>						
demonstratives, such, and the wh-forms	D	Dn	S	S	–	those who came, that which remains: as for few. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	Dn	S	S	–	some that I saw: as above. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	Dn	S	S	–	few who come ever leave: an ordinary postmodifier of the Nom that few heads (I ₁ , I ₂); attaches outside the fused DP, which the inner-phrase restriction states (I ₃ , I ₄). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
cardinals	D	Dn	S	S	–	two that I have seen: as above. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
compound determinatives	D	Dn	S	S	–	anyone who asks, something that you need to know: as above, and ordered after the restrictor (compound family). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
pronouns	L	L	L	L	–	we who have read the report: a restricted range, lexical.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	–	people who came: general.
<i>the compound construction: base permissions and the post-head restrictor (§4.5)</i>						
the general quantifiers	D	D	D	D	1/o/o	the base supplies the compound's premodifier permissions (almost anybody from almost any): inheritance from the base in every account.
compound determinatives	S	S	S	S	4/o/o	the post-head restrictor's position and non-recursion, the exclusion of external determination, and the exclusion of a pre-head adjective are conditions of the compound construction in every account (§4.5); the premodifier permissions inherited from the base (hardly any, hardly anyone) are general in all four.
<i>dependent and independent forms (no/none, my/mine) (§3.4)</i>						
no/none	L	L	L	L	–	inflectional forms of one lexeme selected by the use; the partitive requires none: lexical.
my/mine	L	L	L	L	–	my/mine: as no/none.
pronouns	L	L	L	L	–	she/her: case selected by function.
<i>number agreement and transparency (§2.1)</i>						

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
my/mine	S	S	D	D	2/o/o	Mine is ready / Mine are ready: ordinary Head needs a constructional agreement condition (§4.6); fusion takes the number from the understood nominal (I ₃ , I ₄).
demonstratives, such, and the wh-forms	L	L	L	L	–	this/these agree with the target: the lexeme's inflection.
the general quantifiers	D	D _n	D ₂	D	1/o/o	some of the people were: the transparency rule for quantificational heads, stated once over Nom heads (I ₁) or once over {N, D} heads (I ₂); CGEL states it for the number-transparent nouns and again for the fused-head partitive (I ₃); under I ₄ the rule for nouns covers some. each of the people takes singular agreement by default, a lexical fact in every account. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	D _n	D ₂	D	1/o/o	many of them were: number transparency as for the general quantifiers. Independent a few takes plural agreement from few; peripheral a imposes no singular-count requirement in I ₁ , I ₂ , and I ₄ . CGEL describes the agreement of complex a few in fused-head use. Under I ₂ the derivation admits D as a head of Nom and NP.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	3/o/2	a lot of the work was, a lot of the people were: the number-transparent nouns (CGEL §5.3.3), in every account.
<i>number and grade inflection (§2.3)</i>						
demonstratives, such, and the wh-forms	L	L	L	L	2/o/o	number inflection of two lexemes: lexical in every account; the paradigm parallels nominal number.
the gradable four	L	L	L	L	8/o/o	grade inflection (few/fewer/fewest): lexical; shared with adjectives.
cardinals	L	L	L	L	–	plural cardinals (in threes) are the secondary use (below).
pronouns	L	L	L	L	o/1/o	case and reflexive forms.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	–	count nouns inflect for number: a general rule with lexical exceptions.
proper nouns	L	L	L	L	–	restricted number (the Smiths) is the secondary use (below).
<i>genitive marking (§2.3)</i>						
compound determinatives	D	D ₂	D ₂	D	1/2/o	everyone's, everyone else's: the head-genitive definition, 'the inflection of the head noun' (CGEL p. 479), covers a compound within Noun (I ₁ , I ₄); with a separate D the definition has to be extended to D heads (I ₂ , I ₃), §2.5.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
pronouns	L	L	L	L	–	my/mine, her: suppletive genitive forms.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	1/o/o	dog's, the King of England's: head and phrasal genitives.
proper nouns	D	D	D	D	1/o/o	Kim's, Edward's.
<i>degree modification outside the NP; comparative complements; predicative use (§5.3)</i>						
the articles	D	D _n	D ₂	D	2/o/1	the bigger the better: the as a degree modifier of comparative AdjPs, as much bigger and no better; an NP modifier (I ₁ , I ₂ , I ₄) or a DP modifier (I ₃). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
no/none	D	D _n	D ₂	D	1/o/o	no better: as the bigger. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	D _n	D ₂	D	6/1/o	enough modifying adjectives, adverbs, verbs, and PPs (good enough, quickly enough): an NP modifier under the three NP-projecting implementations, under the permission that covers a great deal smaller and lots better (I ₁ , I ₂ , I ₄); a DP modifier that needs its own permission beside the NP permission (I ₃), §5.3. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	D _n	D ₂	D	4/1/o	much bigger, a lot fewer: as for enough; the comparative complement in more than ten is licensed separately in every account (lexical), and predicative Their enemies were many is a restricted lexical fact. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	13/o/1	a great deal smaller, lots better, heaps worse, three years old: NP modifiers of adjectives in every account (CGEL pp. 549–550).
<i>predeterminer and restricted peripheral modification; the former complex determinatives (§3.2)</i>						
the articles	L	L	S	L	–	a has a restricted peripheral-modifier permission with few, little, and modified many (I ₁ , I ₂ , I ₄), stated on that lexeme rather than supplied by its ordinary Det permission. Its count-singular Det selection applies in many a and such a. I ₃ lists a few, a little, a great many, and many a as complex determinatives.
demonstratives, such, and the wh-forms	D	D _n	D	D	–	such a mess, what a mess: degree such and exclamative what in predeterminer function (I ₁ , I ₂ , I ₄); adjectives in the same function under CGEL (I ₃). The difference is the category, not the machinery. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
the general quantifiers	D	Dn	D	D	2/o/o	all the books, both the copies: a predeterminer modifier realized by the determinative's phrase in every account (a DP in CGEL, an NP here). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	Dn	S	D	o/3/o	a few, a little, a great many: a is a peripheral modifier of the outer NP, while the quantifier heads Det before another nominal and heads the NP in independent use (I ₁ , I ₂ , I ₄). The existing peripheral-attachment rule applies with a restricted lexical permission for a and head-specific internal modifiers; many a has many in predeterminer function on an a-NP. CGEL lists these expressions as complex determinatives (I ₃), §3.2. Under I ₂ the NP projections admit D as a lexical head.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	1/o/o	half a cake, and the fractions: nouns and NPs in predeterminer function in every account (CGEL p. 385 n. 25).
<i>the secondary set-denoting use (§6.1)</i> cardinals	S	S	S	S	–	in threes, in their hundreds: the same rule (I ₁ , I ₄); CGEL's 'secondary use in which they inflect for number and hence belong in the noun category' is a change of category, a second rule (I ₂ , I ₃), §5.5.
compound determinatives	S	S	S	S	–	a special someone, such an anyone, the somebodies: the secondary-use rule of §6.1, one rule over Noun (I ₁ , I ₄); the same rule for nouns plus a conversion rule from D (I ₂ , I ₃ ; CGEL p. 423 n. 43), §6.1 and §5.5.
proper nouns	S	S	S	S	–	an Ophelia, the Smiths: CGEL's own secondary use within Noun (p. 521), the same rule in every account.

5 LIMITS

The grid covers the fragment and selected wider comparisons, including degree modification outside NP structure, comparative complementation, and predicative uses. These entries don't extend the formal fragment specified in the article's §5.1. Interrogative clause structure and coordination remain outside the audit. The independent-*every* cell represents the ordinary dependent pattern; the anaphoric occurrences documented in the quantifier supplement lie outside the fragment.

The grid doesn't score the simple determinatives' genitive restrictions or the agreement and complement conditions needed by the ordinary-Head extension to anaphoric genitives. Nor does it test CGEL's nominal *nylon* and *double* counterexamples, for which the article retains fusion. These limits prevent the cell totals from supplying a complete ranking of the analyses.

The classes are coarse by design; a D under one implementation and a D under another can rest on different primitives (Table 1), and the summary counts should be read with that table. The classifications are contestable at the cell, and the JSON file (`analysis/coverage-audit.json`) is written so that a reader can change one cell and regenerate the tables (`analysis/tools/coverage_audit.py`).

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